Leadership & Consulting in Data Science

Week 6



Team Meeting – Verizon EDA



Create the ***draft*** set of analyses to consider sharing with the client

- Consider importance of insights and relevance to problem
- Determine order…keeping storytelling in mind

Review the EDA generated by each of your team members

Reconvene with the class and compare your key EDA findings with the class

- Create a short (1-3 visuals) from your EDA that tells a story (2-3) minutes
- A team member with the visuals should join the class Zoom session
- Each team will share their story (they do NOT have to be slides)
- Class should be prepared to provide feedback

Week 6 Agenda

1



Measurement and Metrics

2



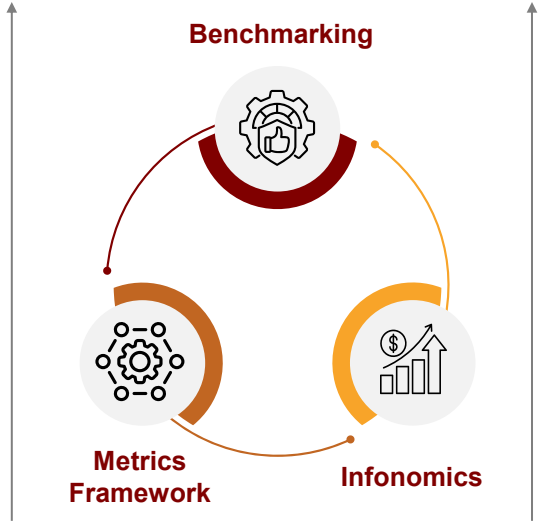
Meeting Management

Metrics capture the progress toward stated vision and business outcomes

Seven Building Block Framework

- Vision
- Strategy
- **Metrics**
- Information Governance
- Organization & Roles
- Information Life Cycle
- Infrastructure

Business Outcomes



- Leading versus lagging indicators
- Metrics “Causality Chain”
- Best Practices

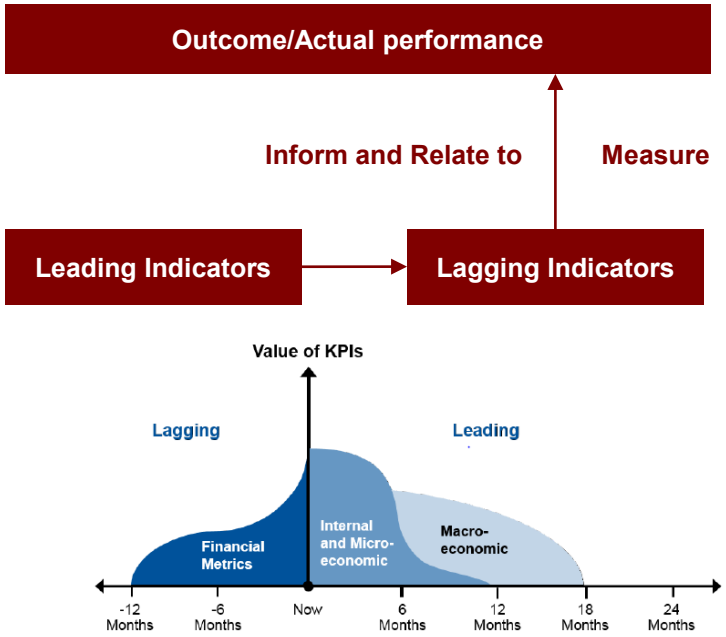
- Evaluate performance relative to industry peers and best-in-class
- Identify opportunities for improvement
- Ensure proper budgeting

- Economic significance to information
- Information valuation frameworks
- Information KPI models

For a Data Science program to succeed, it’s crucial to determine the right scope and role of metrics

Leading and lagging indicators are required to proactively plan, track, manage and measure performance

Causal Relationship



Leading and Lagging Indicators

Lagging indicators are metrics that measure the primary end-state objectives or desired outcomes (typically “backward looking”).

Leading indicators measure activities or events that precede desired outcomes and have a **causal relationship** with specific metrics and financial outcomes. Leading indicators look at more real-time events that contribute to lagging indicators.

Leading indicators measure “how” a proposed business model can be monitored, measured, and monetized. Leading indicators should be used to *hypothesize, assess, test, refine and implement* new business models that have become possible through the advent of new and emerging technologies.

Both leading and lagging indicators are critical to plan, track, manage and measure performance. Linking leading indicators to lagging indicators will improve success and reduce risk.

Leading indicators are part of an organization's competitive advantage, its "secret sauce"

An MIT Sloan Management Review & BCG Global Study (2024)[1]

90% of organizations that use AI to create new KPIs report improvement in their KPIs

34% of organizations currently use AI to create new performance metrics — most still rely on legacy KPIs

60% of business leaders say their KPIs need improvement

Organizations using AI-enhanced KPIs report a 4x increase in employee collaboration and are 2x more likely to achieve greater efficiency than those that do not

Organizations need to have both Lagging and Leading Indicators as part of the overall causality chain. Leading organizations are focused on more real-time events that will influence Lagging Indicators by impacting financial metrics and value creation mechanisms. Leading Indicators will pave the way by providing "early warning signs"

- Reduce costs
- Gain operational efficiencies
- Improve regulatory compliance
- Reduce technical debt

Organizations using AI-enhanced KPIs are 3x more effective at predicting future performance and 3x more likely to see greater financial benefit[1]

1. Source:



Best practices can guide the development and the use of leading and lagging indicators

Metrics should consist of **leading and lagging indicators**. Lagging indicators are used to measure what is being achieved while leading indicators are used to measure how those results will be accomplished

Leading and lagging indicators must be developed by **coordinating efforts** among **cross-functional teams** so that they are common, standard, and understood across the organization

Apply **a causality chain** to build **cause-and-effect relationship** between the chosen drivers of strategic success and outcomes to define, develop, and exploit leading indicators

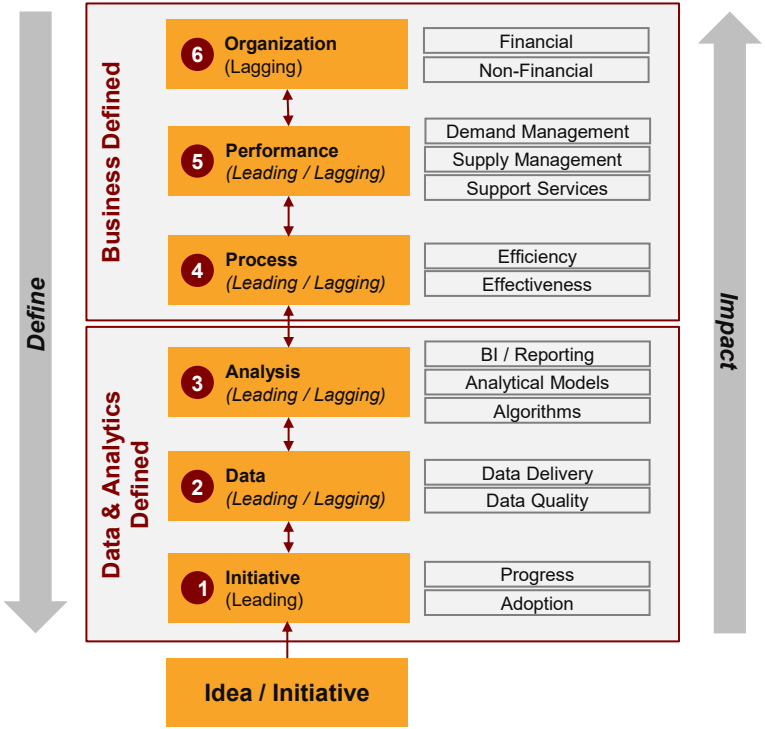
Link leading indicators to lagging indicators as they relate to and inform the lagging indicators

Encourage and propagate **the use of leading indicators** in strategy development, forecasting, and budgeting

Establish standard **metrics definition, baseline**, and set **target**. Ensure **available and quality data** required to capture and measure metrics

Continually revise leading and lagging indicators to remain competitive, relevant and agile

You need to identify your own metrics and work with the business to define the business metrics within the "causality chain"



"Causality Chain" Examples

- Customer retention costs will be decreased by 20%.
- Revenue will increase by 9%.

- Customer satisfaction will be increased by 20%.
- Market share will increase by 5%.

- Returns will be reduced by 60%.

- Market basket analysis confidence increases by 8%

- Customer duplicates will be reduced by 80%.
- Customer address accuracy will be improved by 40%.

- Purchase Data Quality Tool $25K
- 1 FTE, 40 Hours to install and configure the tool

Deploy an address data quality tool and clean up old addresses.

Note: Linkages do NOT have to be numeric or cover all levels

- Users able to answer business questions more quickly
- BI and Analytics teams freed for other tasks

- Increase in the number of analytical models used by the business

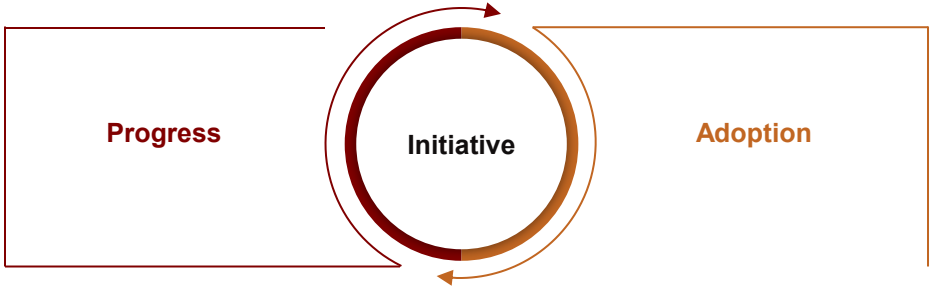
- Users have access to more data and can find it easily

- Purchase licenses for each user
- Install and Deploy Tool
- Train Users

Deploy Self-Serve Analytics

Initiative metrics center on the up-front investment tied to the progress and adoption of ideas and initiatives

1 Initiative



Progress metrics typically focus on

- **Capabilities:** Investing $X in a self-service tool will reduce customer report requests to IT by Y%
- **Tools:** Consolidating visualization tools will cost $X and reduce the number of tools supported from Y to Z and reduce required support personnel from A to B
- **Domains:** Investing $X in a new MDM capability will increase the number of domains supported from Y to Z
- **Skills:** Investing $X in an analytics training module will promote Y people to be self-sufficient advanced analytics users

Adoption metrics typically focus on

- **Business Units:** Implementing a Self-service tool costing $X will provide analytics capabilities to Y business units
- **Functions:** Implementing a data science tool costing $X will provide advanced analytics capabilities across Y functions
- **Systems:** Investing $X in enhanced data integration capabilities will consolidate data across Y systems and provide shared availability to W functions and Z business units

Data Quality measures consist of subjective and objective quality measures

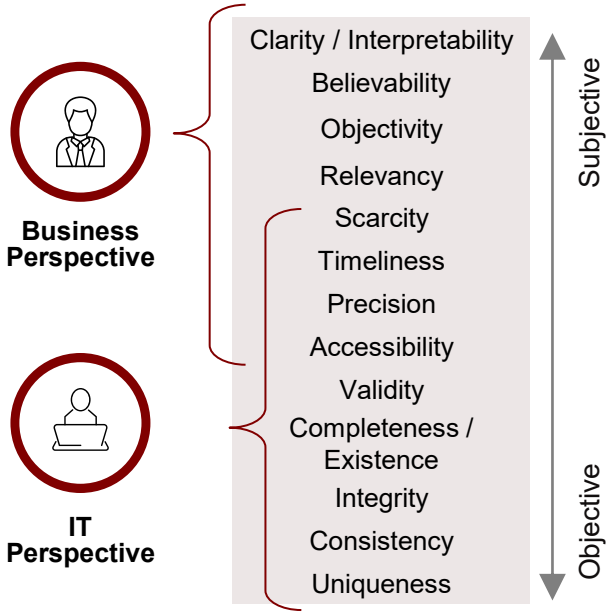
Business and IT stakeholders hold different views regarding data quality

- **Businesspeople perceive data quality in terms of subjective concepts,** such as trust, confidence and understandability of the data
- **IT personnel tend to focus on objective terms that can be measured directly,** such as validity of data and consistency of data across systems

Both perspectives are valid, but neither is useful in isolation when seeking consensus and establishing objectives for a data quality improvement initiative

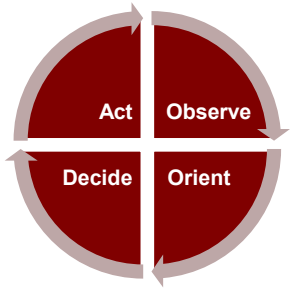
At the same time, both measures are co-dependent.

- We cannot expect to improve the perception or subjective measures of data quality unless we make objective improvements
- The appropriate objective metrics and thresholds are best informed by the subjective needs of the business



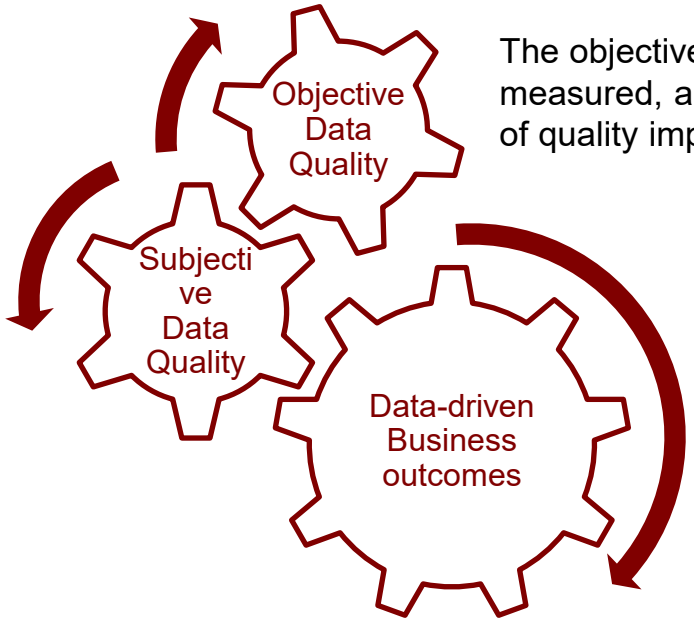
A Data Governance initiative motivates continuous data quality improvement informed by metrics

2 Data



OODA Loop

Perception of data quality typically lags underlying objective quality improvements. User perception can be measured through surveys



The objective data quality baseline can be measured, and improvements tracked as a result of quality improvement initiatives

Increased user confidence leads to more effective decision making and processes for greater reliability and efficiency of operation

Defining Data Quality measures and measurement tools

2 Data

Survey perception | Tool-based query | Third party verification

Attribute	Description	Measurement
Usability	Business process(es) and/or individuals understand and are able to leverage data	Survey perception
Clarity	Data has a unique, defined meaning and can be easily interpreted	Survey perception
Believability	Data is deemed credible by those using it	Survey perception
Objectivity	Data is unbiased and impartial and not dependent on the judgment, interpretation or evaluation of individuals	Survey perception
Relevancy	The data is applicable to one or more business process or decision	Survey perception
Scarcity	The data is proprietary, secret and difficult to come by	Tool-based query, Survey perception
Timeliness	Data is updated and available in time to meet business requirements and OLA	Tool-based query, Survey perception
Precision	Data is recorded with the precision required by business processes or decision	Tool-based query, Survey perception
Accessibility	Data is readily retrieved and/or integrated into business processes or decision	Tool-based query, Survey perception
Validity	Data accurately represents reality, possibly qualified by a verifiable source	Tool-based query, Third party verification
Uniqueness	Data that exists in multiple places has the same value hence disambiguated	Tool-based query, Third party verification
Completeness/Existence	Records are not missing fields and datasets are not missing instances	Tool-based query, Third party verification
Integrity	The appropriate links and relationships exist among data	Tool-based query
Consistency	Data that exists in multiple locations is similarly represented and/or structured	Tool-based query

Data Quality consists of twelve steps that start with candidate KPIs (leading) and Key Result Indicators (lagging)

2 Data

1. Identify candidate KPIs and Key Result Indicators
2. Apply initial data profiling
3. Design and implement dashboard
4. Determine data quality goals
5. Define data standards
6. Extend external trust-based model
7. Establish data quality measures as input to data governance
8. Delegate data quality actions to data stewards
9. Establish enterprise data quality team
10. Establish a data quality review as a release "stage gate"
11. Communicate the data quality benefits
12. Leverage external/industry peer groups

Iterative

Delivery metrics ensure users receive the right data at the right time with an understanding for how to interpret it

2 Data

Attribute	Question to Answer	Sample Leading Metric	Sample Lagging Metric
Existence	Can I get it?	• Percentage of records in existence by domain (complete records/total records)	• Number of missing or estimated data records by domain • Number of complete data records by domain
Discovery	Can I find it?	• Percentage of business terms in information catalog	• Number of reports pulled per month
Access	Can I get to it?	• Percentage of total to potential users • Number of new reports/metrics created	• Number of reports pulled per month • Total users per month
Understandability	Can I interpret it?	• Validation score (using 1-5 Likert scale)	• Number of reports retired per month
Fitness	Can I use it?	• Records rejected or flagged	• Relevancy score by data domain • (data domain applicability to business processes/total business processes)

Supports users answering their questions

Analytics Type	Question to Answer
Descriptive	What happened?
Diagnostic	Why did it happen?
Predictive	What will happen?
Prescriptive	What should we do?

Data Quality and Delivery support analytics used by the business to meet their goals and objectives

3 Analysis

Analytics Type	Question to Answer
Descriptive	What happened?
Diagnostic	Why did it happen?
Predictive	What will happen?
Prescriptive	What should we do?

These questions are answered via:

- BI/Reports
- Analytical Models
- Algorithms

Measured & Improved by

Metric Type	Sample Metrics
Performance	• Model accuracy • Percentage of statistically significant models • Time-to-deliver models/reports
Usage	• Model output adoption rate • Number of models supported • Number of real-time models supported • Number of BI reports supported • Number of users requesting new data/reports
Users	• Number of analysts supported • Number of data scientists supported • Number of new users per month

Process metrics drive better decisions across the processes being changed or impacted

4 Process

Four Best Practices to Design Meaningful Metrics That Drive Better Decisions for BPM

Clarify the business outcomes that need to be achieved

Determine the value drivers that influence your business outcomes

Use value drivers to understand the factors that influence business outcomes

Select meaningful metrics that drive well-informed decision making

Ask the following questions to measure Business Process outcomes:

- What are the most critical processes for business success?
- Why are we making the change?
- What business outcomes will this BPM project support?
- What key stakeholders will be impacted?
- What questions need to be answered?
- What decisions need to be made?
- What aspects of the organization need to differ, and which should remain the same?
- Which processes will be affected?

Attribute	Sample Leading Metrics	Sample Lagging Metric
Customer Experience	• On-Time Order Delivery • Order Accuracy • Service Performance • Material Quality • SLA Effectiveness • Customer Care Performance	• NA
Campaign Effectiveness	• NA	• Click Through Rate • Outbound / Inbound Sales Calls • Lead Lifecycle • Social Media Interactions
Workplace Efficiency	• Units Produced Per Hour • Product Defects • Handling Time • Revenue Per FTE	• NA
Speed of Innovation	• Time-to-Market • Number of New Patents	• % Revenue from products introduced • Actual vs. Targeted Break-Even Time
Operational Excellence	• Productivity • Gross Margin	• Customer Satisfaction Score • Employee Satisfaction Score

Capability metrics measure performance across demand, supply and support services

5 Performance

Business Aspect	Attribute	Sample Metric			
Demand Management	Market Responsiveness	Target Market Index	Market Coverage Index	Market Share Index	Opportunity/Threat Index
		Product Portfolio Index	Channel Profitability Index	Configurability Index	
	Sales Effectiveness	Sales Opportunity Index	Sales Cycle Index	Sales Close Index	Sales Price Index
		Cost-of-Sales Index	Forecast Accuracy	Customer Retention Index	
	Product Development Effectiveness	New Product Index	Feature Function Index	Time-to-Market Index	R&D Success Index
Supply Management	Customer Responsiveness	On-Time Delivery	Order Fill Rate	Material Quality	Service Accuracy
		Service Performance	Customer Care Performance	Agreement Effectiveness	Transformation Ratio
	Supplier Effectiveness	Supplier On-Time Delivery	Supplier Order Fill Rate	Supplier Material Quality	Supplier Service Accuracy
		Supplier Service Performance	Supplier Care Performance	Supplier Agreement Effectiveness	Supplier Transformation Ratio
	Operational Efficiency	Cash-to-Cash Cycle Time	Conversion Cost	Asset Utilization	Sigma Value
Support Services	Human Resources Responsiveness	Recruitment Effectiveness Index	Benefits Administration Index	Skills Inventory Index	Employee Training Index
		HR Advisory Index	HR Total Cost Index		
	Information Technology Responsiveness	System Performance	IT Support Performance	Partnership Ratio	Service-Level Effectiveness
		New Project Index	IT Total Cost Index		
	Finance and Regulatory Responsiveness	Compliance Index	Accuracy Index	Advisory Index	Cost-of-Service Index

Business performance metrics disaggregate organizational metrics into attributes that can be more clearly monitored and managed

Organization metrics leverage existing business measures and KPIs that are of critical importance to the organization

6 Organization

Financial

- **Income statement line items**
  - Gross Income
  - Operating Expense
  - Net Income
- **Balance sheet line items**
  - Total Assets
  - Total Liabilities & Owner's Equity
  - Total Current Liabilities
- **Financial Performance**
  - Return on Equity
  - Earnings per Share
  - Gross Profit Margin

Non-Financial

- **Employee**
  - Wellness
  - Satisfaction
- **Product**
  - Active users
  - Quality
- **Customer**
  - Share of Wallet
  - Lifetime Value
- **Market**
  - Market Share
  - Demand
- **Competitor**
  - Share of Voice
  - Keyword Ranking
- **Corporate Responsibility**
  - Environmental footprint
  - Charitable contributions

Isolating a Data and Analytics Program's contribution to an organizational metric can be very challenging and therefore not undertaken by many organizations, but **it can be done**:

1. Identify organizational metrics from executive/organizational scorecards
2. Identify the key operational outcomes and their underlying processes that contribute to those organizational metrics
3. Design data and analytics metrics to link into these key business processes

Week 6 Agenda

1

Measurement and Metrics

2



Meeting Management

Agenda: managing meetings from decision to follow-through

1	**Decide**	Is a meeting the right tool — and if so, which format?
2	**Plan**	Purpose, agenda, participants, materials, and roles
3	**Facilitate**	Opening, timeboxing, participation, and a clean close
4	**Follow through**	Minutes, decisions, owners, deadlines, and evaluation
5	**Apply**	Common pitfalls, their counters, and a reusable checklist

Not every conversation needs to be a meeting

Hold a meeting when…

- A decision needs input from several people at once
- The topic is ambiguous, sensitive, or likely to spark debate
- You need to build alignment, trust, or shared context
- Live back-and-forth will be faster than a written thread
- The work is creative and benefits from real-time collaboration

Use another channel when…

- You are broadcasting information people can read on their own
- A status update would work just as well as a written summary
- Only one or two people are genuinely involved
- The decision is already made and simply needs communicating
- Participants need time to think before they can respond well

Rule of thumb: if there is no decision to make, no problem to solve, and no relationship to build, cancel the meeting and send a note instead.

Managing a meeting effectively involves careful planning and execution







Match the format, length, and guest list to the purpose

Decision

PURPOSE
Choose between options and commit

WHO
The decision owner plus 3–5 informed voices

LENGTH
30 minutes

DONE WHEN
A decision, an owner, and a date exist

Working session

PURPOSE
Build or solve something together

WHO
3–6 people doing the actual work

LENGTH
60–90 minutes

DONE WHEN
A draft or artifact exists

Status review

PURPOSE
Surface risks, blockers, and trade-offs

WHO
The delivery team and its sponsor

LENGTH
15–25 minutes

DONE WHEN
Escalations have named owners

Brainstorm

PURPOSE
Generate a wide range of options

WHO
5–8 people with diverse perspectives

LENGTH
45 minutes

DONE WHEN
A ranked shortlist exists

Default to the shortest format that achieves the purpose — and retire any recurring meeting that no longer has one.

Pre-meeting planning is critical to an effective meeting



Define the Meeting Purpose and Objectives

- Clearly state the purpose of the meeting
- Set specific objectives and outcomes you want to achieve

Create an Agenda

- Develop a detailed agenda outlining topics, discussion points, and time allocations
- Share the agenda with participants in advance

Select Participants

- Identify the key stakeholders and participants necessary for the meeting
- Ensure their availability and send out invitations

Choose a Date, Time, and Location

- Confirm availability and book the room or meeting link

Prepare Materials

- Gather and distribute any pre-read materials or documents relevant to the meeting

Assign Roles

- Appoint a facilitator, timekeeper, note-taker, and any other necessary roles

Clear roles keep the meeting on track

1 Facilitator

Runs the agenda, opens and closes the meeting, keeps discussion on topic, and makes sure every voice is heard.

2 Timekeeper

Tracks the clock against the agenda, flags when an item runs long, and prompts the group to decide or defer.

3 Note-taker

Captures decisions, action items, owners, and deadlines — not a transcript — and circulates the minutes.

4 Decision owner

Frames the options, makes the call when the group cannot, and is accountable for the outcome afterward.

5 Contributors

Read the pre-reads, come prepared, argue from evidence rather than opinion, and own their action items.

6 Subject-matter expert

Joins for one specific agenda item, briefs the group, then is released — not required for the full meeting.

During the meeting, you need to stay on top of progress



Opening

- Welcome participants and provide a brief overview of the agenda and objectives

Stick to the Agenda

- Follow the agenda and adjust as needed to stay on schedule

Engage Participants

- Encourage active participation and contributions from all attendees
- Draw out quieter participants and test for dissenting views before moving on

Document Key Points

- Assign a note-taker to capture key points, decisions, and action items

Facilitate Discussion

- Keep the discussion focused and ensure that all viewpoints are heard
- Manage any disruptions or off-topic discussions – maintain a “parking lot” for any topics that are important, but off-topic

Close the Meeting

- Summarize key takeaways, decisions, action items (responsibility and deadline), and next steps
- Thank participants for their time and contributions

Most meeting failures are predictable — and preventable

COMMON PITFALL	HOW TO COUNTER IT
No clear purpose — the meeting exists because it is on the calendar	State the objective and the decision to be made in the invitation. If you cannot write one, cancel.
Wrong people in the room — the decision-maker is missing, or ten optional attendees are not	Invite by role, not by habit. Mark attendance optional where it is, and send notes to everyone else.
Discussion drifts and the agenda runs out of time	Timebox every item, name a timekeeper, and move off-topic items to the parking lot.
A few voices dominate while quieter participants never weigh in	Ask for input by name, use round-robins, and invite written comments before or after the meeting.
The meeting ends with no decisions, owners, or deadlines	Close with a two-minute recap: what was decided, who owns what, and by when.

Post-meeting execution is critical to delivering value



Distribute Meeting Minutes

- Send within 24 hours while the discussion is still fresh
- Lead with decisions and action items, not a transcript

Communicate Outcomes

- Share meeting decisions and outcomes with relevant stakeholders

Follow-Up

- Ensure action items are being worked on and delivered as planned, sending reminders or updates as needed

Archive Meeting Materials

- Organize and store meeting materials for future reference

Evaluate Meeting Effectiveness

- Assess whether the meeting achieved its objectives
- Collect feedback from participants to improve future meetings
- Run an After Action Report (AAR) for high-stakes meetings

A working checklist you can reuse for every meeting

BEFORE

- ☐ Purpose and objective written in one sentence
- ☐ Agenda with topics, owners, and time allocations
- ☐ Only the people who need to be there are invited
- ☐ Pre-reads sent at least 24 hours ahead
- ☐ Facilitator, timekeeper, and note-taker assigned
- ☐ Date, time, and location or link confirmed

DURING

- ☐ Open with purpose, agenda, and desired outcomes
- ☐ Hold each item to its timebox
- ☐ Draw out every participant, not just the loudest
- ☐ Park off-topic items rather than chasing them
- ☐ Capture decisions, actions, owners, and deadlines
- ☐ Close with a recap and confirm next steps

AFTER

- ☐ Minutes distributed within 24 hours
- ☐ Decisions shared with stakeholders who were absent
- ☐ Action items tracked with owners and due dates
- ☐ Reminders sent ahead of the deadlines
- ☐ Materials archived where the team can find them
- ☐ Feedback collected; run an AAR if warranted

Keep this beside your invitation template and run through it before you hit send.

Verizon Case Preparation (1 of 3)



The value of a data science model to a business organization is *not* measured by:

- Parameter significance
- Classification accuracy or F1 score
- Silhouette scores and other cluster separation metrics

Instead, the value of a data science model is measured by:

- Profit gained over current status quo
- Losses mitigated or hedged
- Change in risk exposure
- Change in cash flow
- Growth in market share
- Derivative metrics such as customer satisfaction, new accounts opened, or shortenings of the average hiring timeline

Verizon Case Preparation (2 of 3)



As a telecommunication provider, Verizon offers both long-term service contracts and the devices used to access those services. Data science models could help it to secure more potential profits and prevent more avoidable losses.

		Verizon Action	
		Offer contract	*Deny contract*
Customer Action	*Pay all bills*	Profit occurs	Profit avoided
	Default early	Loss occurs	Loss avoided

Break down the value of the data science model both by directly measuring the associated profits or losses as well as comparing to these new values to the status quo. *For example,*

$$\Delta Value = \Delta Profit - \Delta Loss$$
$$= \Delta(Paying\ Customers) \cdot (Customer\ LTV) + \Delta(Defaults\ Avoided) \cdot (Cost\ of\ Default)$$
$$- \Delta(Defaulting\ Customers) \cdot (Cost\ of\ Default) - \Delta(Safe\ Applicants\ Denied) \cdot (Customer\ LTV)$$

The formula above may be theoretically correct, but you may not have the pieces you need, or another value calculation may be a better fit!

Verizon Case Preparation (3 of 3)



Meet with your team for to discuss and do the following:

1. Identify which inputs you need to estimate the value of your solution
   a) Inputs which have been supplied to you
   b) Inputs you can learn from the data,
   c) Inputs you can learn from outside sources,
   d) Inputs you can estimate, *and*
   e) Inputs which are totally unknown
2. Iterate on #1 above until you have a feasible way forward
3. Present your pathway to estimating value back to the class, **using at most 1 minute and 2 slides.** (This is not a client-facing presentation.) Cite the sources you use and describe how you will acquire the information you don't yet have. **Stay succinct.** Questions may follow.